

NAVONILA KARMAKAR

Final Year BS-MS Physics Student

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RESEARCH EXPERIENCE

MS Thesis Project at Max Planck Institute for Astrophysics, Germany *06/2026-Present*

Supervisors: **Prof. Dr. Volker Springel** and **Dr. Alankar Dutta**

Hydrodynamic cloud-crushing simulation using PLUTO: Extending classical cloud-crushing (cold gas clouds of CGM in hot galactic wind) setup in two ways: (1) adding wind divergence/expansion, and (2) fractal, multi-clump cloud complex instead of single spherical cloud, and analysing how each affects the cloud survival, evolution and morphology, and the physics behind, ongoing work on cloud analysis.

Independent Study at IISER Kolkata; Supervisor: **Prof. Rajesh Kumble Nayak** *08/2026-Present*
Semester project on **Bayesian inference and parameter estimation of gravitational waves.**

Remote Project Intern at ESO, Germany; Supervisor: **Dr. Morten Andersen** *05/2025-Present*

Characterizing the intermediate and low-mass stellar content of Barbá 2 cluster: imaging data of H and Ks bands from Gemini FLAMINGOS2, data reduction using DRAGONS pipeline and Astropy, infrared photometric (aperture) study of cluster stars using photutils, calibration with 2MASS photometric data, completeness limit determination from radially binned CMD and corresponding luminosity function, estimation of stellar masses from individual dereddening, and finally derivation of cluster mass and stellar number density profile.

Astronomy Lab Course Semester Project, IISER Kolkata *04/2026-05/2026*

Structural and Morphological Analysis of Galaxies using Image Statistics: Data acquisition from SDSS SkyServer for three different types of galaxies: NGC 2967 (Sc spiral), NGC 3351 (SBb barred spiral), NGC 3379 (E1 elliptical), image processing, surface brightness profile and model fitting, error and noise analysis, CAS morphological parameters determination, isophote analysis, global fourier analysis (2D FFT), radial Fourier analysis (Annular 1D FFT), correlation and statistical analysis, radial structural decomposition, final comparison and classification. ([Report](#))

LIGO GET-GO Summer Intern at ICG, University of Portsmouth, UK *05/2025-08/2025*

Supervisors: **Prof. Dr. Laura Nuttall** and **Dr. Rahul Dhurkunde**

Mitigating the effect of glitches in the search of precessing systems: basics of GW searches such as matched filtering, signal consistency tests, coincident detection, ranking statistic, candidate significance etc using PyCBC; differences in waveform, template bank and ranking statistic for precessing searches; mathematical formalism of harmonic decomposition of precessing waveforms, performing precessing searches on a few days of data of O3 in CIT cluster, identification of GPS times of glitches that are affecting precessing searches, types of glitches, estimation of sensitivity for a simulated population of signals (injections) at a given FAR, comparing the sensitivity of the new model with existing general searches. ([Presentation](#))

Summer Research Fellow at ARIES Nainital, India; Supervisor: **Dr. Saurabh** *05/2024-07/2024*

Understanding Star Formation in Young Clusters from Ground-based Observation: observation in 3.6 m Devasthal Optical Telescope, imaging and spectroscopy with TANSPEC; data reduction using IRAF, UVBRI photometric (PSF) study of NGC 2316 stars using DAOPHOT II, calibration with Landolt's standard SA 98 to determine true magnitude of stars, 2MASS near-infrared photometric data study for cluster identification, kinematic study from GAIA proper motion data for determining membership probability, astrometry using IRAF, GAIA Tool & WCSTools, isochrone fitting with intrinsic Zero-Age Main Sequence on HR Diagram of cluster stars for morphological analysis. ([Report](#))

Summer Intern at **CESSI, IISER Kolkata**; Supervisor: **Dr. Sudip Kumar Garain** 07/2023
Literature review on **Stellar Evolution: Interstellar Cloud to Black Holes**. ([Report](#))

EDUCATION

Indian Institute of Science Education and Research (IISER), Kolkata 10/2022 - *Present*
BS-MS Dual Degree, Major in Physics CGPA: 8.0
Coursework: classical mechanics, quantum mechanics, mathematical physics, thermodynamics and statistical physics, electrodynamics, computational physics, electronics theory & lab, nuclear physics lab, introductory astrophysics, general relativity, quantum field theory, condensed matter theory & lab, introductory cosmology, astronomy lab, fluid dynamics and MHD, atomic & optical physics, advanced optics theory & lab.

Bishnupur High School 08/2020 - 06/2022
West Bengal Council of Higher Secondary Education(WBCHSE) Percentage: 93.8

Shibdas Central Girls' High School 01/2014 - 07/2020
West Bengal Board of Secondary Education(WBBSE) Percentage: 93.8

PUBLICATIONS

Fragmented in the flow: survival of fractal cloud complexes in expanding galactic outflows
Navonila Karmakar, Alankar Dutta et al., 2027 (In preparation)

The infrared view of the Galactic star cluster Barba 2: The low and intermediate mass content
M. Andersen, N. Karmakar, S. Millstone et al., 2026 (In preparation)

SKILLS

Programming Languages and Softwares: Python (SciPy, AstroPy, Photutils, h5py, PyCBC, Bilby), C/C++, Julia, MATLAB, L^AT_EX, IRAF, DS9, DAOPHOT II, GAIA Tool, OriginLab, Anaconda, Git, PLUTO, ParaView, GNUPlot, MESA, CASA, HEASoft.

Languages: English, Bengali, Hindi

SUMMER & WINTER SCHOOLS/WORKSHOPS/CONFERENCES

IMPRS-HD Summer School, University of Heidelberg 09/2026
Cosmic Multiplicity: Evolution and Fate of Binary Stars

PLUTO/OpenGadget3 Symposium 06/2026
Max Planck Institute for Astronomy, Heidelberg, Germany

CMF2IMF - The Origin of the Stellar Initial Mass Function 06/2026
European Southern Observatory, Garching, Germany

Cosmology, Astrophysics & Astronomy 03/2026
Special Lecture Series by International Leading Experts, Kolkata

Beyond LIGO-India: Deci-Hz GW Space Mission, IISER Kolkata 03/2026

Glorious Women Online Symposium, India (Online)	03/2026
LIGO GET-GO Student Conference, University of Glasgow, UK (Presentation)	07/2025
SOKENDAI Asian Winter School, NAOJ & SOKENDAI, Japan	02/2025
Sagan Summer Workshop, NASA Exoplanet Science Institute, Caltech	07/2024
Lecture Series on Physics of Inflation & Reheating, Raman Research Institute	07/2024
Summer School on Gravitational-Wave Astronomy, ICTS Bangalore	07/2024
REYES student (virtual), University of California	06/2024-07/2024
Summer School in Astronomy & Astrophysics, IUCAA Pune	05/2024-06/2024
Workshop on Gravity, Cosmology and Raychaudhuri's Equation, IUCAA & JU	03/2024
18th Kavli Asian Winter School on Strings, Particles and Cosmology Yukawa Institute for Theoretical Physics and ICTP (online attendee)	12/2023
10th IAPT National Student Symposium on Physics, Chandigarh	08/2023
START (Space Science and Technology Awareness Training), ISRO	07/2023-08/2023

HONOURS AND AWARDS

MPG-IISER MS Project Research Fellow Fellowship sponsored by Max Planck Institute for Astrophysics , selected through MPA Internship program and MPG-IISER program	06/2026-Present
LIGO GET-GO Summer Research Fellow Gravitational wave detector Expertise and Technology-GrOwth (GET-GO) fellowship sponsored by University of Glasgow , selected through LIGO SURF program	05/2025-08/2025
IASc-INSa-NASI Summer Research Fellow Summer Research Fellowship Program (SRFP) sponsored by Indian Academy of Sciences (Bangalore), Indian National Science Academy (New Delhi), National Academy of Sciences, India (Prayagraj)	05/2024-07/2024
Silver Honour (Youth), IAAC (International Astronomy and Astrophysics Competition)	06/2023
Bronze Medalist (Senior) , National Level Astrophysics Quiz Competition IAPT and A.P.J. Abdul Kalam National Council of Young Scientists	09/2021